

Interview Preparation Notes

Q: Explain your project without technologies

I built an AI-powered Resume Analyzer. A student uploads their resume, and the system automatically reads it, checks how strong it is, identifies what skills they have, and tells them which companies they are a good fit for. It also shows what is missing in their resume, gives personalized recommendations to improve it, generates interview questions based on their skills, and downloads a full PDF report. The goal is to help freshers like me prepare better for placements.

Q: What technologies did you use?

For the backend I used Python with Flask framework. For the frontend I used HTML, CSS, Bootstrap 5, and Chart.js for charts. For the database I used SQLite. For text extraction from PDF and DOCX files I used pdfplumber and python-docx. For JD matching I used scikit-learn's TF-IDF vectorizer with cosine similarity. For PDF report generation I used ReportLab. For authentication I used Flask sessions with SHA-256 password hashing.

Q: What challenges did you face in the project?

One major challenge was handling different resume formats because every resume structure is different. Sometimes extracted text was not properly aligned. Another challenge was database errors during authentication and report generation.

Q: How is data stored in your project?

User details, uploaded resume information, analysis results, and report metadata are stored in SQLite tables. Passwords are never stored in plain text — they are hashed using SHA-256 before saving.

Q: What database did you use and why?

I used SQLite. I chose it because it is lightweight, requires no separate server setup, and is built into Python. In future I would migrate to MySQL or PostgreSQL for better scalability and concurrent user handling.

Q: What do your teammates feel about you?

My teammates feel that I am hardworking, reliable, and supportive. They know that I take responsibility seriously and try to complete my work on time. Whenever someone faces a technical issue, I help them patiently and work together to solve it. They also feel comfortable working with me because I communicate well, listen to others' ideas, and stay calm even during stressful situations.

Q: Tell me about a failure in your life

Earlier, I was mainly focused only on academics and getting good marks. Later, I realized that academics alone are not enough for the software industry. So I started concentrating more on practical learning, coding, projects, and problem-solving. That shift helped me improve my technical skills and build real-world projects.

Q: What is your most successful achievement?

One of my biggest achievements was becoming an academic topper in my second year. My college recognized my performance and awarded me with a scholarship reward. That achievement gave me a lot of confidence in my learning ability. Later, I realized that academics alone are not enough for the software industry, so I started focusing more on practical skills, coding, and real-world projects. That journey helped me successfully build my AI Resume Analyzer project from scratch.

Q: Middle tier / mid layer

Flask — app.py. It connects frontend and database.

Q: UI layer

HTML, Bootstrap 5, Jinja2 templates, Chart.js for charts.

Q: Why did you use Flask over Django or Node.js?

Flask is a micro-framework that gives more control and is less opinionated compared to Django. For a project of this scale, Django's ORM and admin overhead were unnecessary. Flask allowed us to build a modular architecture where each feature is a separate module. Node.js was not chosen because Python has superior NLP and data processing libraries like scikit-learn, which were central to our project.

Q: How ATS score is calculated

The ATS score is calculated out of 100 based on four main factors: 1. Presence of important resume sections like Education, Experience, Skills, Projects, and Summary. 2. Detection of technical skills. 3. Use of action verbs. 4. Keyword density. Each factor is weighted, and the combined score is normalized to 100.

Q: How do you store resume data in SQLite?

When a user uploads a resume, we store the filename, upload timestamp, extracted text, detected skills list, and the calculated ATS score in a SQLite database table.

Q: Did you face any conflicts or disagreements with your teammates? How did you handle it?

Yes, we had a minor disagreement about which database to use — one team member preferred MySQL while I preferred SQLite for simplicity. Instead of arguing, we discussed the trade-offs objectively. We agreed on SQLite for development with the understanding we'd switch to MySQL for production scaling.

Q: What projects have you built using the technologies you learned?

I built an AI-based Resume Analyzer using Python and Flask. In this project, users can upload their resumes, and the system automatically extracts the text from the resume, analyzes the skills mentioned, compares them with the relevant job requirements, and provides actionable suggestions to improve the resume and increase the chances of getting shortlisted.

Q: What makes a good team member?

A good team member communicates clearly, completes tasks on time, supports teammates, and is willing to learn.

Q: What if your teammate is not completing work?

I would first talk politely with them and understand the issue. If needed, I would help them or inform the team so the project work does not get delayed.

Q: Do you prefer working alone or in a team?

I can work in both situations, but teamwork helps in sharing ideas and completing projects faster.

Q: How did your team manage the project?

We divided tasks based on skills. We discussed progress regularly and combined all modules step by step.

Q: Did you help your teammates?

Yes, whenever someone had difficulty understanding backend logic or Flask concepts, I tried to explain and support them.
